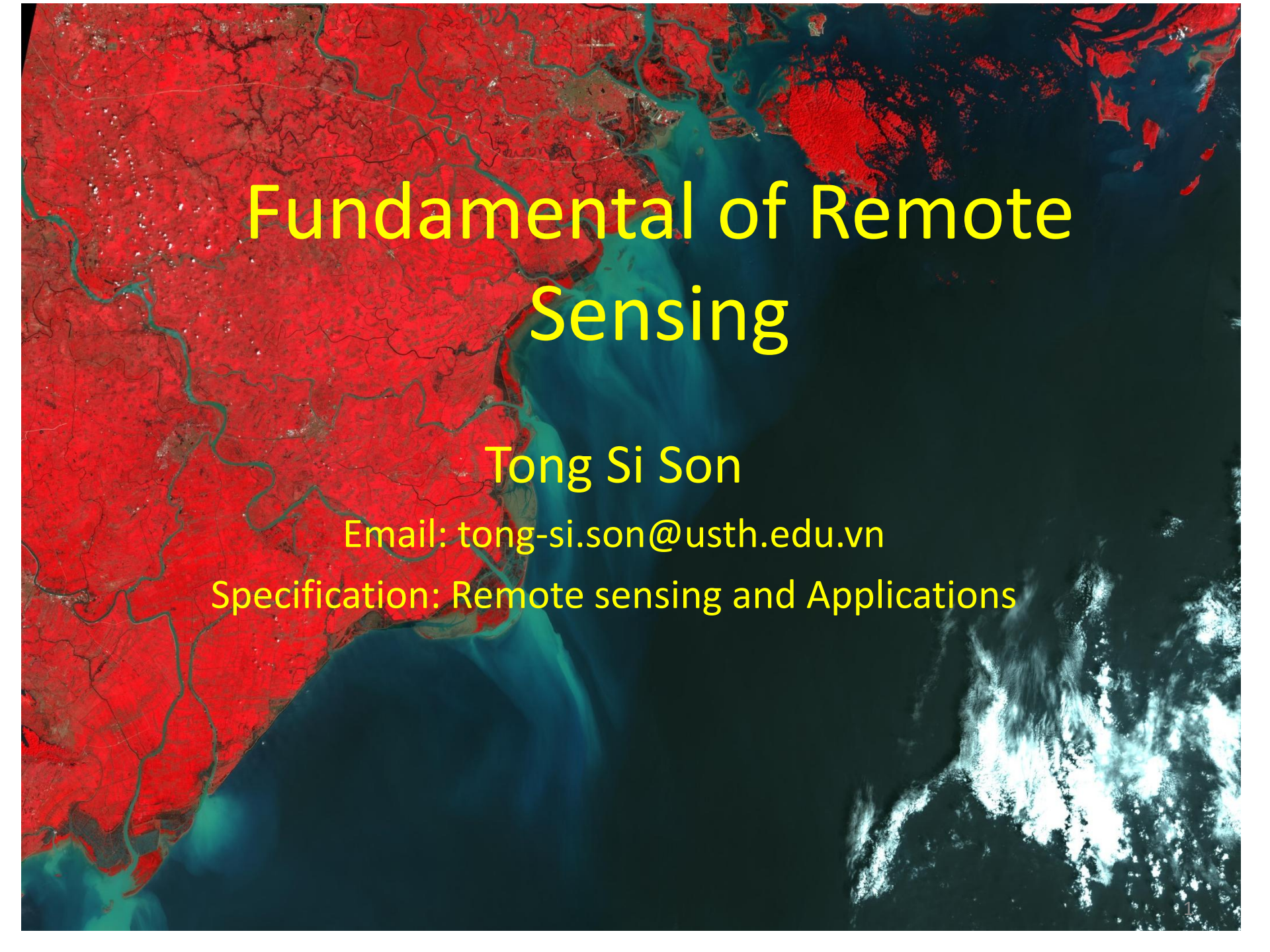


A satellite image of a coastal region. The land is colored red, showing a complex network of rivers and waterways. The surrounding water is dark blue. In the bottom right corner, there is a bright, white, cloud-like feature. The text is overlaid on the image in a yellow, sans-serif font.

Fundamental of Remote Sensing

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Specification: Remote sensing and Applications

Practices

1. Exploring satellite data
2. Practice to measure the reflectance of surface object in the lab
3. Image pre-processing
4. Extract Information from satellite data
5. Midterm presentation

1. Exploring satellite data

1/ Download data and software

ENVI, SNAP, OTB, ERDAS, IDRISI, GEOMATICA, ECOGNITION, Google Earth Engine..

2/ Explore the satellite data

3/ Color composition

4/ Define objects on the satellite data and investigate the spectral reflectance

5/ Calculate sensor spectral radial

Exploring satellite data

Top Free Satellite Image Sources

1. **USGS** EarthExplorer / GloVis

Landsat, ASTER, MODIS (30 m)



2. **Copernicus** Copernicus Sentinel Program

Sentinel-1, Sentinel-2 (10 m)



3. **NASA** NASA Earthdata / Worldview

MODIS, VIIRS, SMAP



4. **Google Earth Engine**

Cloud-based Analysis



5. **EOS** EOS LandViewer

Landsat, Sentinel, NDVI



6. **JAXA** JAXA G-Portal

ALOS, ALOS-2 (Radar)



7. **ISRO** ISRO Bhuvan

Resourcesat, Cartosat



8. **China** China GEO Data Cloud

Gaofen Series



9. **MAXAR** Maxar Open Data

High Resolution (<1 m)



10. **UNOSAT** (United Nations)

Disaster & Emergency



FREE Satellite Imagery for GIS & Remote Sensing

USGS – EarthExplorer (Landsat Data)

<https://earthexplorer.usgs.gov>

Copernicus Data Space (Sentinel Satellites)

<https://dataspace.copernicus.eu>

NASA Earthdata Search

<https://search.earthdata.nasa.gov>

Google Earth Engine

<https://earthengine.google.com>

EOS LandViewer

<https://eos.com/landviewer>

JAXA G-Portal

<https://gportal.jaxa.jp>

ISRO Bhuvan Geoportal

<https://bhuvan.nrsc.gov.in>

China GEO Data Cloud

<http://www.gscloud.cn>

Maxar Open Data Program

<https://www.maxar.com/open-data>

UNOSAT (United Nations Satellite Centre)

<https://unosat.org>

1.1 Download data and software

Download satellite image from

<https://browser.dataspace.copernicus.eu/>
<https://earthexplorer.usgs.gov/>

Download SNAP for processing satellite image

<https://step.esa.int/main/download/snap-download/>

	Windows 64-Bit	Windows 32-Bit	Mac OS X	Unix 64-bit
Sentinel Toolboxes	These installers contain the Sentinel-1 , Sentinel-2 , Sentinel-3 Toolboxes, download size is close to 900MB.			
	Main Download	Main Download	Main Download	Main Download
	Mirror Download	Mirror Download	Mirror Download	Mirror Download
SMOS Toolbox	These installers contain only the SMOS Toolbox , download size is close to 500MB. Download also the Format Conversion Tool (Earth Explorer to NetCDF) and the user manual .			
	Main Download	Main Download	Main Download	Main Download
	Mirror Download	Mirror Download	Mirror Download	Mirror Download
All Toolboxes	These installers contain the Sentinel-1 , Sentinel-2 , Sentinel-3 Toolboxes, SMOS and PROBA-V Toolbox, download size is close to 1GB.			
	Main Download	Main Download	Main Download	Main Download
	Mirror Download	Mirror Download	Mirror Download	Mirror Download

1.2 Exploring satellite data

Requirement:

Download a Landsat 8 image; Level 2, covering Hanoi

Download a Hyperion image; Level 1T

Download and install SNAP software for processing images.

- Read information of metadata file

Structure of images

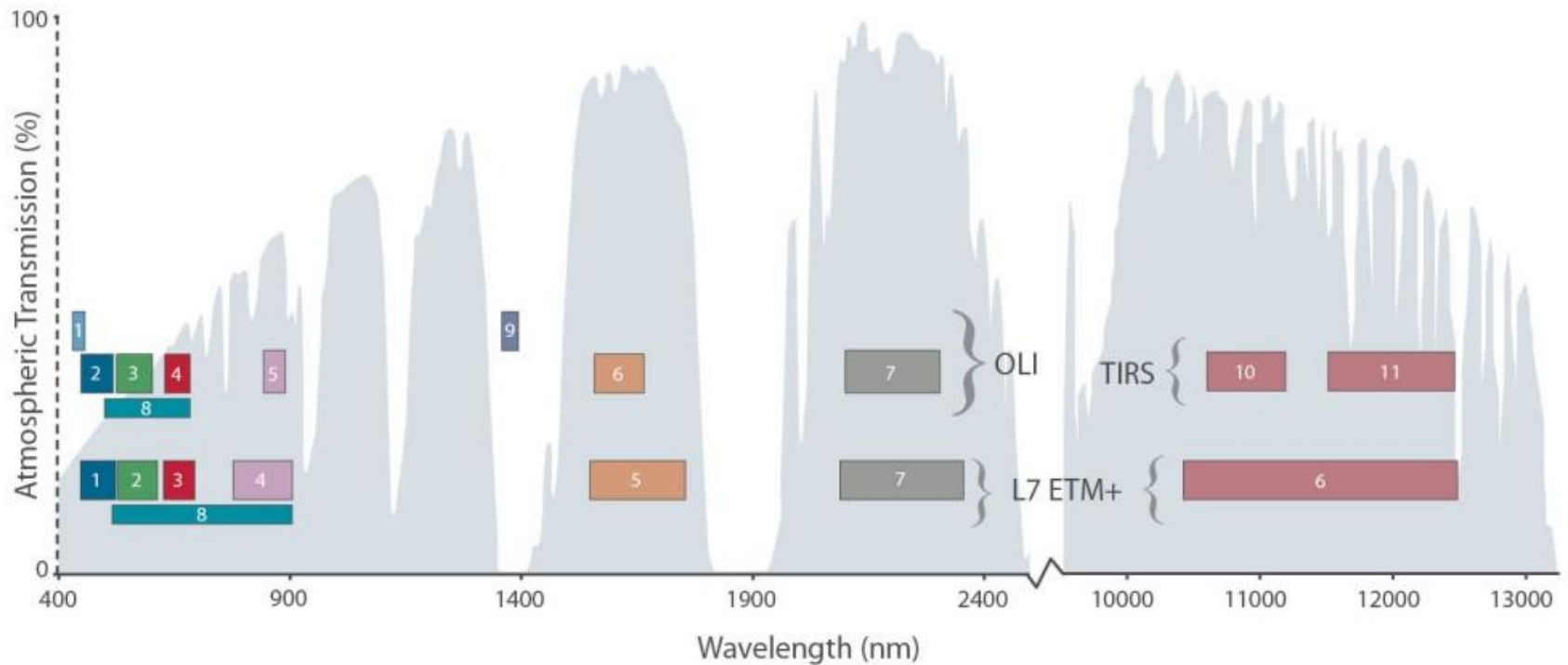
Row/Path, Coordinates, Value of image

Metadata information

Type of sensor, acquisition time, gain value...

- Open images

Landsat 7, 8



Characteristic of Landsat image

Landsat-7 ETM+ Bands (μm)			Landsat-8 OLI and <i>TIRS</i> Bands (μm)		
			30 m Coastal/Aerosol	0.435 - 0.451	Band 1
Band 1	30 m Blue	0.441 - 0.514	30 m Blue	0.452 - 0.512	Band 2
Band 2	30 m Green	0.519 - 0.601	30 m Green	0.533 - 0.590	Band 3
Band 3	30 m Red	0.631 - 0.692	30 m Red	0.636 - 0.673	Band 4
Band 4	30 m NIR	0.772 - 0.898	30 m NIR	0.851 - 0.879	Band 5
Band 5	30 m SWIR-1	1.547 - 1.749	30 m SWIR-1	1.566 - 1.651	Band 6
Band 6	60 m TIR	10.31 - 12.36	<i>100 m TIR-1</i>	<i>10.60 – 11.19</i>	Band 10
			<i>100 m TIR-2</i>	<i>11.50 – 12.51</i>	Band 11
Band 7	30 m SWIR-2	2.064 - 2.345	30 m SWIR-2	2.107 - 2.294	Band 7
Band 8	15 m Pan	0.515 - 0.896	15 m Pan	0.503 - 0.676	Band 8
			30 m Cirrus	1.363 - 1.384	Band 9

Hyperion image

TABLE 1 PARAMETERS OF HYPERION

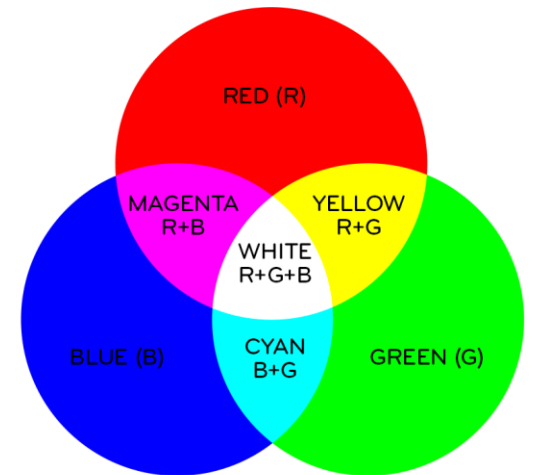
Instrument	EO-1/Hyperion
Spectral Range	0.4-2.5 microns
Visible Bands	35
Near infrared Bands	35
Short Wave infrared	172
Spatial Resolution	30 m
Swath Width	7.5 km
Spectral Coverage	Continuous
Number of Bands	220
Temporal Resolution	200 days



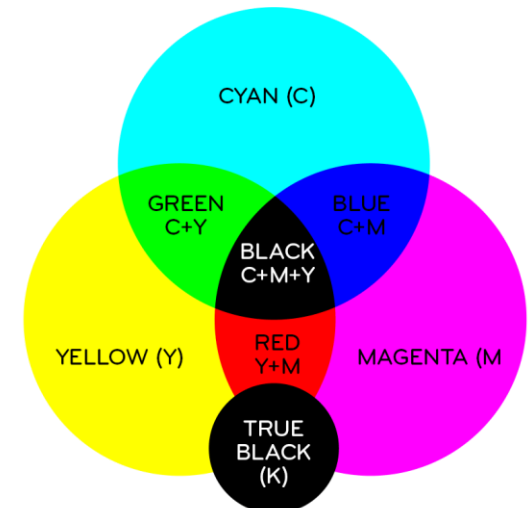
Hyperion Band	Wavelength (nm)	Approx.FWHM (nm)	Spatial resolution (m)
Band 1 - 8	355 - 426	11.38	30
Band 8 -17	436 - 518	11.38	30
Band 17 - 27	528 - 620	11.0	30
Band 28 - 37	630 - 721	10.6	30
Band 38 - 47	732 - 823	10.7	30
Band 48 - 140	833 - 1548	11	30
Band 141- 185	1558-2002	10.9	30
Band 185 - 242	2012 -2577	10.4	30

1.3 Color composition

- Gray scale: 0-255 (for individual band)
- Assign Band 4, Band 3, Band 2 for RGB → Natural color composition
- Assign Band 5, Band 4, Band 3 for RGB → Pseudo color composition

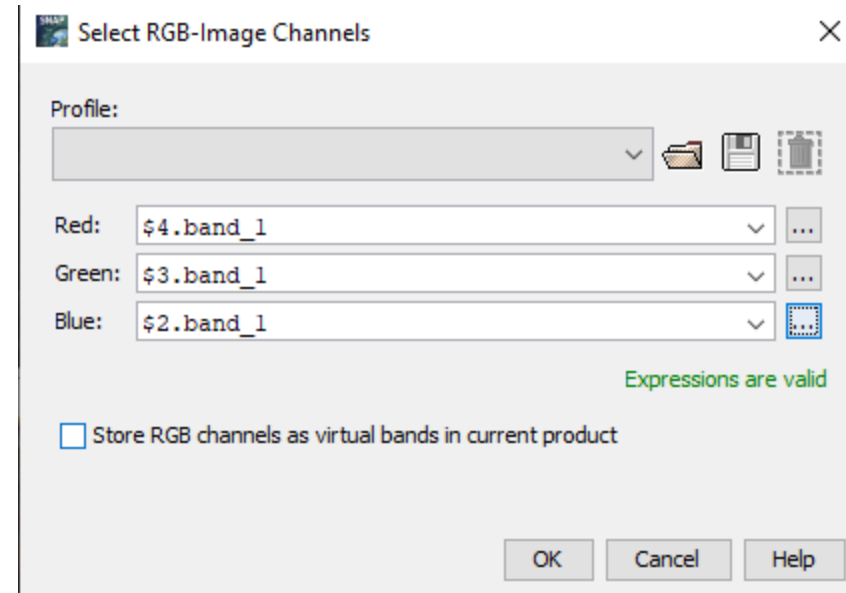
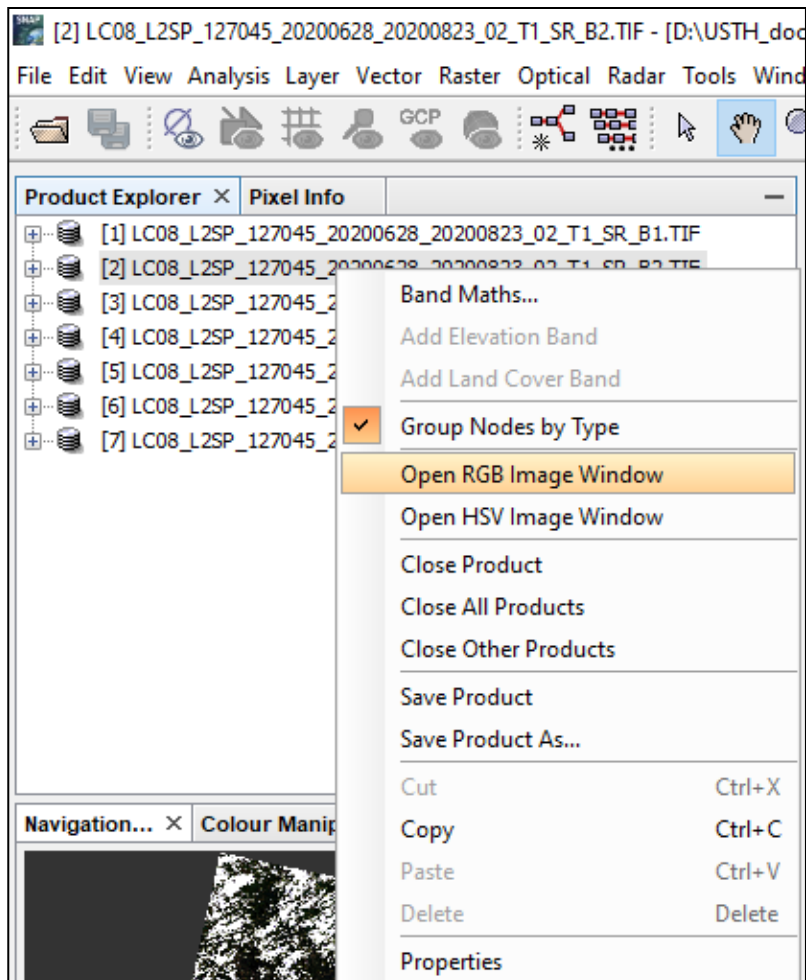


RGB COLOR SPACE (ADDITIVE COLOR)
PRIMARIES FOR LIGHT, OR ONSCREEN COLORS



CMYK COLOR SPACE (SUBTRACTIVE COLOR)
PRIMARIES FOR 4-COLOR PRINTING

Color composition in SNAP



1.4 Spectral distribution

- Vector/ New vector data container
- Analysis/ Profile plot
- Analysis/ Histogram
- Calculate band ratio

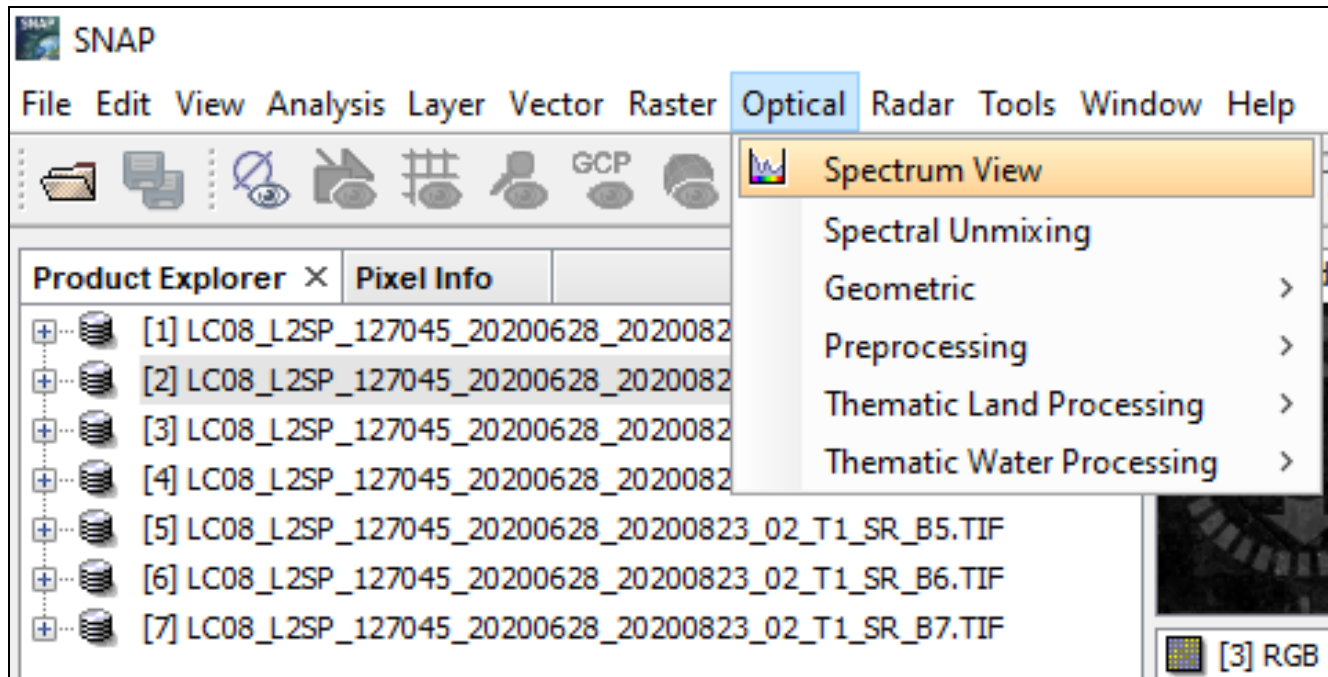
NDWI (Normalization Different Water index)

NDVI (Normalization Different Vegetation index)

1.5 Define surface object and investigate the spectral reflectance

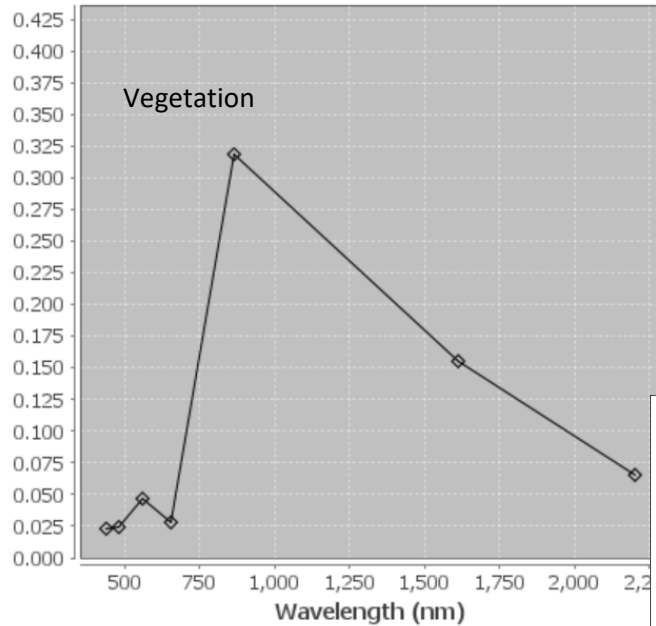
- Investigate the spectral reflectance: Using Spectrum view tool
- Compare the spectral reflectance with the theory

Spectrum view tool

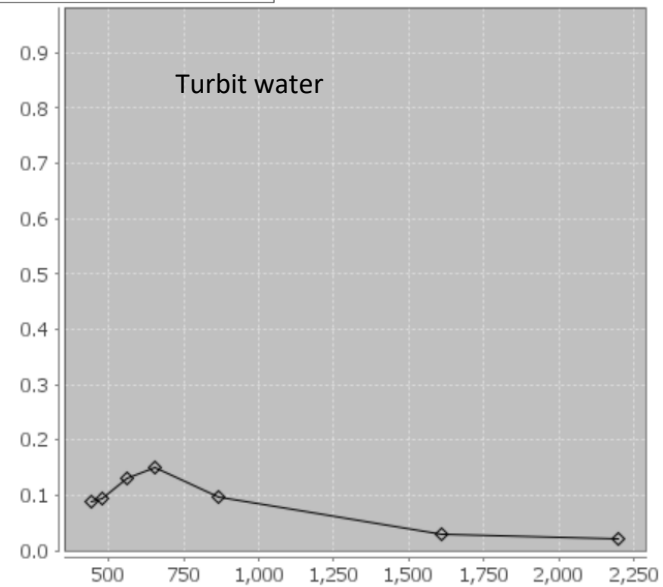
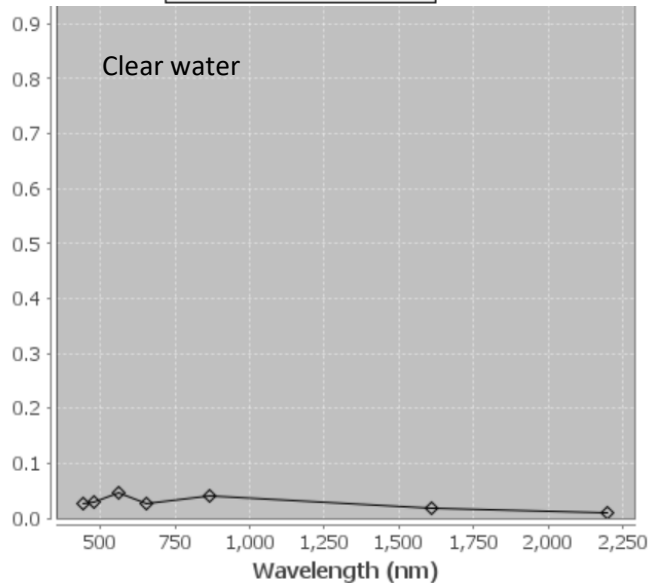
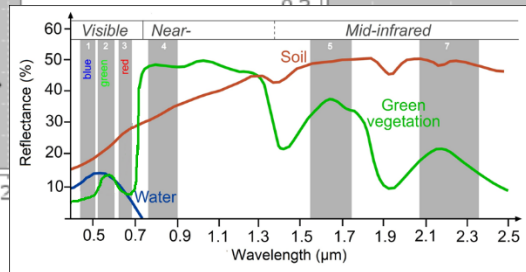
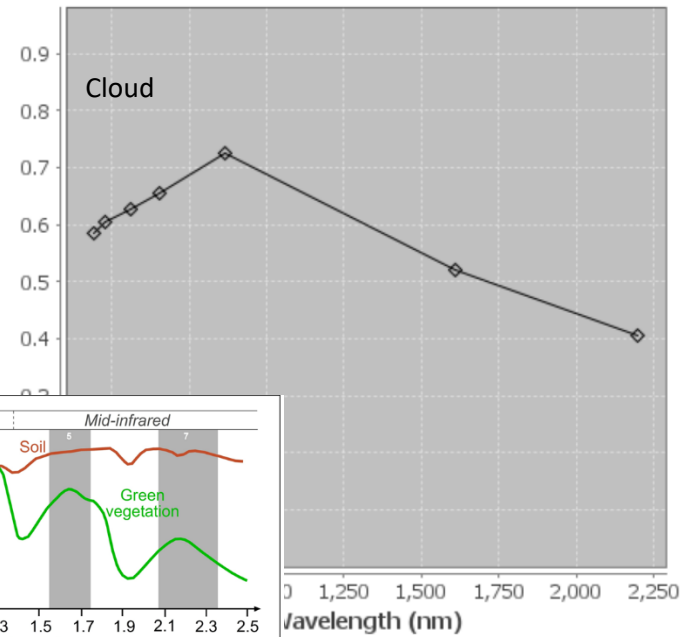


Reflectance

Spectrum View



—◇— Bands measured in



Thank you for your attention!